

Character Frequency in Heap

You are given a string S consisting of only uppercase English letters (A to Z) and a number k . Your task is to print the top k characters based on their frequency in the string. The character with a higher frequency has a higher priority, and if frequencies are equal, characters with a higher alphabetic value ($Z > A$) have higher priority.

Input Format

- The first line contains an integer k , the number of characters to print.
- The second line contains a string S , consisting only of uppercase English letters

Output Format

Display k characters and their frequency with the following format: `jcharacterj jfrequencyj`.

Constraints

- $1 \leq k \leq 26$
- $5 \leq |S| \leq 100$
- Implement your algorithm in C using Heap concept for insertion and removal.

Sample Input 1 (standard input)

```
3
ABBCCDDDD
```

Sample Output 1 (standard output)

```
D 3
C 3
B 2
```

Explanation 1

From the input string ABBCCCD, we compute the frequencies A: 1, B: 2, C: 3, D: 3. Characters C and D both appear 3 times. Since D has a higher alphabetic value than C, D is prioritized.

Only the top $k = 3$ characters are displayed. Then, we get: D (3), C (3), B (2).

Sample Input 2 (standard input)

```
20
BINUSUNIVERSITY
```

Sample Output 2 (standard output)

```
I 3
U 2
S 2
N 2
Y 1
V 1
T 1
R 1
E 1
B 1
```

Explanation 2

Since the input string consists of only 10 unique characters, all characters are displayed. If the frequencies are the same, prioritize the character with the higher alphabetical value (in descending order).